

DEVELOPER + INVESTOR COLLABORATION BRIEF

Build the career evidence layer for data and AI talent.

DataPath connects a curated skills graph, technical assessment, practical evidence and career execution in one personalised workspace.

18	28	252	9
career paths	skill areas	curated topics	industry lenses

CURRENT STAGE / FUNCTIONAL PRODUCT BUILD / PRE-PUBLIC VALIDATION

Career learning is fragmented

Learners move between course libraries, static roadmaps, quiz sites, project lists, interview tools and spreadsheets. The missing layer is a trusted system that connects the target role to learning, practice, evidence and career action.

The user problem

What should I learn for this role? Which gaps matter?
How do I practise them? What proves I can do the work? What should I revisit before an interview?

The DataPath answer

A curated role-skill-topic graph becomes a personalised plan. Each topic connects to guidance, practice, assessment, evidence, industry context and career preparation.

Initial audience

Students, career switchers and working professionals pursuing analytics, data engineering, governance, data science and AI roles. The first geographic advantage is relevance to Egypt and the wider MENA market, while the core product model remains globally applicable.

Why now

Data and AI career paths are expanding faster than most learners can evaluate curricula. Employers still need observable skill evidence, while learners need a practical bridge from study to interview.

Product principle

AI personalises and explains against the approved knowledge model. It does not invent the curriculum, prerequisites, assessment rules or evidence standard.

Category ambition

Become the workspace where a learner plans a role transition, builds verified capability and carries that evidence into applications and interviews.

Trust position

Readiness is derived from visible work: completed requirements, saved evidence, technical checks and skill exams. Self-report remains clearly labelled.

A connected career workspace

The current build is more than a concept. It implements the complete navigation and core learning logic needed to validate the product journey.

Personalised onboarding

Role discovery, 18 paths, experience, interests, free or paid resources, target period, weekly time and industry selection.

Curated knowledge graph

28 skill areas and 252 levelled topics with stable identifiers, prerequisites, estimates, official resources and learning guidance.

Assessment hierarchy

Lesson quizzes, level checks, skill exams and cumulative reviews with weak-topic remediation.

Evidence system

Lesson notes, project records, completion checks, skill status and an exportable Proof Ledger.

Career execution

Portfolio guidance, interview studio, CV preparation, job-readiness checklists and application tracking.

User controls

Dark and light themes, responsive navigation, session logging, profile export/import, guest storage and optional accounts.

Industry intelligence

Nine sector lenses add vocabulary, workflows, metrics, risks and project direction to the roadmap.

Optional AI coach

Provider-based assistance stays behind an explicit user action and receives only bounded curriculum context.

Current status

The application runs locally and passes type checks, automated domain tests and a production build. Public deployment infrastructure, live user validation and outcome data are the next stage; no traction figures are claimed in this brief.

The learning-to-employment loop

A strong session always answers: what should I do now, why does it matter, how do I practise it and what evidence proves I can do it?

01 ORIENT	Select a role and industry; understand fit and real work.
02 DIAGNOSE	Combine self-assessment with optional technical checks.
03 LEARN	Use authored guidance and approved external resources.
04 PRACTISE	Complete a practical task, case or runnable exercise.
05 VALIDATE	Pass topic, level, skill and cumulative assessments.
06 REPAIR	Return weak topics to the roadmap with clear remediation.
07 APPLY	Build a sector-relevant portfolio case and explain decisions.
08 PROVE	Carry transparent evidence into CVs, interviews and applications.

Defensibility grows through use

The durable asset is the connection between a versioned curriculum graph, technical attempts, remediation history, project evidence, sector cases and career outcomes. Content breadth alone is easy to copy; a trusted evidence standard and outcome dataset are harder.

Free first; monetise validated value

The complete current experience is free. Monetisation should begin only after activation, learning outcomes and retention are demonstrated.

PHASE	OFFER	CUSTOMER / REVENUE	GUARDRAIL
Now	Complete learning and career workspace	Free access to build trust and outcome evidence	No artificial feature locks
Consumer Plus	Runnable labs, richer feedback, mentor review, advanced mock interviews	Monthly or annual learner subscription	Core curriculum and proof remain accessible
Institutions	Cohorts, assignments, instructor view and readiness analytics	University and training-centre licences	Consent and transparent measures
Employers	Talent challenges, verified profiles and role-specific screening	Recruitment partnerships and challenge fees	No opaque candidate scoring

Initial wedge

Build one exceptional end-to-end path: Data Analyst in Banking, with deep lessons, executable SQL, a sector case, technical assessment and reviewed portfolio output.

Expansion logic

Prove the loop in one role and sector, then reuse platform capabilities while expanding authored content into adjacent roles and industries.

Distribution

Start with learner communities, universities, training providers and practitioners who can contribute cases, mentoring and credibility.

What must be measured

Activation, useful weekly sessions, assessment improvement, evidence produced, interview progress and learner-reported career outcomes.

A practical foundation for iteration

The product uses shared deterministic learning logic so the roadmap, assessments and evidence rules remain consistent across the client and server.

CLIENT	SHARED DOMAIN	SERVER	PERSISTENCE
React 19 Vite React Query / tRPC Responsive CSS Local guest state	Zod state schema 18 roles 28 skills / 252 topics Planner + prerequisites Quiz + evidence rules	Express tRPC routers Authentication AI boundary Validation + rate limits	Drizzle ORM MySQL when configured Owner-scoped records Optimistic revisions JSON import / export

Security foundation

Signed HttpOnly sessions, scrypt password handling when configured, same-origin write checks, Zod bounds, security headers, rate limits and user-scoped records.

Privacy stance

Guest data stays in browser storage. AI sharing requires a deliberate user action. Export and import keep the learner's state portable.

Highest-value engineering work

Runnable SQL and Python environments, production identity/database operations, assessment analytics, adaptive scheduling and public proof profiles.

Architecture principle

Keep curriculum and assessment policy deterministic; use AI for explanation, feedback and gap prioritisation within explicit boundaries.

Milestones that reduce risk

Each phase should produce evidence for the next investment decision rather than adding breadth without proof.

0-90 DAYS / Activation + quality

Deeply author the Data Analyst path; prototype a SQL sandbox; instrument onboarding and first-session funnels; improve focused-session modes; configure production identity, database and recovery.

3-6 MONTHS / Evidence + retention

Add Python execution; adaptive review scheduling; privacy-controlled proof profiles; mentor or peer project review; validate weekly retention and assessment recovery.

6-12 MONTHS / Distribution + revenue

Pilot university cohorts; add employer-authored challenges; test Consumer Plus for labs and review; form practitioner partnerships; publish an outcome study.

Decision gates

Gate 1 / usability

Can a new learner select a goal and complete a meaningful first mission without guidance?

Gate 2 / learning

Do technical attempts and remediation show measurable improvement?

Gate 3 / evidence

Do learners produce project artifacts and proof they are willing to share?

Gate 4 / demand

Will learners or institutions pay for execution, review and cohort value after the free core works?

Where partners can make the difference

DataPath is looking for collaborators who can turn a strong functional foundation into a validated, public product with credible learning outcomes.

Developer and expert contribution

For product engineers

Own production deployment, identity, persistence, observability, analytics and the secure execution environment for SQL and Python.

For learning engineers

Build assessment authoring tools, testable exercises, adaptive review and content-versioning workflows.

For data practitioners

Review role maps, author realistic cases and question banks, and define evidence that hiring teams respect.

For product and design partners

Refine onboarding, focused sessions, proof profiles and the path from learning activity to career action.

Investor and strategic contribution

For angel and pre-seed partners

Support a focused validation phase covering infrastructure, authored content, practitioner review and learner acquisition.

For strategic investors

Open access to universities, training organisations, employers and regional practitioner networks.

Capital use priorities

Product and engineering; expert curriculum development; secure lab infrastructure; user research and analytics; early distribution.

The near-term outcome

A deployed Data Analyst in Banking experience with observable activation, learning improvement, evidence creation and retention metrics.

What remains to prove

The product has a broad functional base. The next stage is about depth, reliability and evidence of demand.

Learning depth

Many paths require deeper expert-authored explanations, examples, misconceptions, exercises and question banks.

Practical validation

Runnable work and external project review are needed to strengthen proof beyond quizzes and self-recorded evidence.

Production readiness

Public launch needs deployed identity, database operations, recovery, monitoring, backups and incident processes.

Outcome evidence

The team must establish user cohorts and measure learning improvement, retention and career progress.

Commercial proof

Pricing and institutional demand remain hypotheses until learner value and repeat use are established.

Focus risk

The broad taxonomy must not distract from making one path exceptionally useful before expanding.

The collaboration proposal

Start with a focused working session around the Data Analyst in Banking wedge. Review the live build, agree on the first validation cohort, define the 90-day milestones and decide which product, technical, content or capital contribution each partner can own.

DataPath is currently a functional pre-public product. This brief is for discussion and collaboration; it does not present audited financial projections, certified learning claims or verified market traction.